

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <signal.h>

// Signal handler to catch Ctrl+C (SIGINT)
void handle_sigint(int sig) {
    printf("\nInterrupt signal (%d) detected. Exiting gracefully.\n", sig);
    exit(0);
}

int main() {
    char input[256];

    // Register the signal handler for Ctrl+C
    signal(SIGINT, handle_sigint);

    printf("=====\n");
    printf("      C Interactive Shell Initialized      \n");
    printf(" Type 'help' for options, or 'exit' to quit.\n");
    printf("=====\n");

    // 1. Create Main Loop
    while (1) {
        // Display Prompt
        printf("app> ");

        // Read User Input & Handle EOF (Ctrl+D)
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nEOF received. Exiting application. Goodbye!\n");
            break; // Exit the loop on Ctrl+D
        }

        // Remove the trailing newline character added by fgets
        input[strcspn(input, "\n")] = '\0';

        // Skip processing if input is empty
        if (strlen(input) == 0) {
            continue;
        }

        // 2. Handle Exit Conditions
        if (strcmp(input, "exit") == 0 || strcmp(input, "quit") == 0 || strcmp(input, "q") == 0) {
            printf("Exiting application. Goodbye!\n");
            break; // Break the control flow loop
        }

        // Handle other commands
        else if (strcmp(input, "help") == 0) {
            printf("Available Commands:\n");
            printf("  help - Display this help message\n");
            printf("  exit - Quit the interactive application\n");
        }
        else {
            printf("Unknown command: '%s'. Type 'help' for commands.\n", input);
        }
    }
}
```

Output:

```
LST@LAPTOP-EK5DCJVE:~# nano skill3_1.c
LST@LAPTOP-EK5DCJVE:~# gcc skill3_1.c -o skill3_1
LST@LAPTOP-EK5DCJVE:~# ./skill3_1
=====
      C Interactive Shell Initialized
    Type 'help' for options, or 'exit' to quit.
=====
app> help
Available Commands:
  help - Display this help message
  exit - Quit the interactive application
app> test_command
Unknown command: 'test_command'. Type 'help' for commands.
app> exit
Exiting application. Goodbye!
LST@LAPTOP-EK5DCJVE:~# |
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. Open()
2. Read()
3. Write()
4. Close()

open()

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    char buffer[100];
    int source, destination, bytes;
    // open() - Opens File1 for reading
    source = open("File1.txt", O_RDONLY);
    // open() - Opens File2 for writing
    destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    // read() and write() are used inside the loop
    while((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }
    // close() - Closes both files
    close(source);
    close(destination);
    printf("File copied successfully.\n");
    return 0;
}
```

Output:

```
LST@LAPTOP-EK5DCJVE:~# nano skill3_2.c
LST@LAPTOP-EK5DCJVE:~# nano file1.txt
LST@LAPTOP-EK5DCJVE:~# nano file2.txt
LST@LAPTOP-EK5DCJVE:~# gcc skill3_2.c -o skill3_2
LST@LAPTOP-EK5DCJVE:~# ./skill3_2
File copied successfully.
LST@LAPTOP-EK5DCJVE:~# |
```